

OM SHUKLA

Software Engineer || Green Card Holder || No Visa Sponsorship Required
Bettendorf, Iowa, United States

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Summary

Results-driven Software Engineer with a Bachelor of Science in Software Engineering from Iowa State University. Experienced in Agile development, problem-solving, and collaborative team environments. Proficient in Java, JavaScript, C, C++, data structures, MySQL, mobile application development (Android Studio), and game development (Unreal Engine). Skilled in designing and implementing scalable solutions to meet IT business needs. Green Card holder (no visa sponsorship required) and open to relocation.

Technical Skills

Languages: Java, JavaScript, C, C++, MySQL

Frontend: React.js, Tailwind CSS, HTML5, CSS3, JavaScript

Backend: Spring Boot, REST APIs, JPA, MySQL

Tools & Technologies: Git, GitHub, Android Studio, Unreal Engine 5, Verilog, Intel Quartus Prime, Unity, Blender, Discord, Trello

Experience

RUMBLE - Outdoor Roomba

Jan 2025 – Dec 2025

Iowa State University

Ames, IA

- Designed and developed responsive React web application for managing autonomous outdoor cleaning robots with secure user authentication and real-time monitoring.
- Implemented user registration, login, password recovery, profile management, and form validation with error handling.
- Built interactive dashboard with live robot tracking via Google Maps API, fleet statistics visualization, and robot assignment management.
- Developed settings page for profile management, password changes, and account preference updates.
- Implemented user interface features including animations, loading states, and accessibility standards.
- Built REST API endpoints for user management and robot assignment operations using Spring Boot and Java.
- Tech Stack: React.js, JavaScript, Tailwind CSS, Spring Boot, Java, REST APIs, Google Maps API, Git

Repair Technician

Jan 2024 – May 2024

Iowa State University Book Store

Ames, IA

- Diagnosed and repaired hardware and software issues on Apple and Dell devices including laptops and desktops.
- Performed component replacements including displays, motherboards, batteries, and keyboards.
- Maintained detailed repair records and documented customer issues for quality assurance.

RSFS (Ridiculously Simple File System)

Apr 2024 – Apr 2024

Iowa State University

Ames, IA

- Developed a complete in-memory file system in C, implementing both basic and advanced file operations while mastering concurrent programming challenges.
- Built all 8 core API functions (open, append, read, fseek, close, delete, write, cut) supporting multi-block file storage with bitmap-based resource allocation.
- Solved the readers-writers concurrency problem using POSIX pthreads, allowing multiple readers simultaneous access while enforcing exclusive writer locks to prevent data corruption.
- Designed a complete file system architecture, including linked-list directory management, inode allocation tracking, and synchronized access controls across the entire system.
- Validated implementation thoroughly with sequential tests covering file operations across block boundaries and concurrent stress tests with simultaneous readers and writers to verify correctness under contention.

Card Shark – Android Card Game

Aug 2022 – Dec 2022

Iowa State University

Ames, IA

- Implemented REST APIs to support gameplay logic and deck mechanics for multiplayer card game application.
- Designed relational database schema for player cards, decks, and game lobbies using SQL and MySQL.
- Developed inventory logic, battle mechanics, and game state management using Spring Boot and Java.
- Tech Stack: Spring Boot, Java, JPA, SQL, MySQL, GitHub

Sales Floor Associate - TechCyte

Jun 2022 – Sep 2022

Iowa State University Book Store

Ames, IA

- Recommended computer devices and technical solutions to customers by analyzing their needs and requirements.
- Maintained floor displays, organized inventory stocks, and ensured product availability in retail store.
- Provided technical assistance and support to students and staff using troubleshooting and customer service skills.

Assignment Tracker Web App

Jan 2022 – May 2022

Iowa State University

Ames, IA

- Developed responsive user interface using HTML5, CSS, and JavaScript for web application.
- Built authentication system, admin tools, and assignment management screens with comprehensive form validation.

Traffic Light System

Jan 2022 – Apr 2022

Iowa State University

Ames, IA

- Designed traffic control system using Verilog hardware description language and block design tools.
- Implemented system on DE2-115 FPGA board using Intel Quartus Prime, digital logic, and hardware integration.

Pokémon Game

Jan 2022 – Apr 2022

Iowa State University

Ames, IA

- Implemented Pokémon statistics, attributes, and game maps using C programming language.
- Developed game battles, combat mechanics, and text-based user interface using C.

Text-Based Maze Runner Game

Jan 2021 – May 2021

Iowa State University

Ames, IA

- Developed text-based maze game in C using file input/output, pointer manipulation, and data structures.
- Collaborated with team members to debug issues, troubleshoot problems, and implement game features.

2D Chess Game

Jan 2021 – May 2021

Iowa State University

Ames, IA

- Developed graphical chess game using Java programming language with Swing GUI framework.
- Implemented chess rules, move validation algorithms, and game logic components.

Android Developer Intern

Jan 2020 – May 2020

John Deere Academy

Bettendorf, IA

- Designed and built Android application user interface with intuitive navigation and user experience flows.
- Implemented application logic and functionality while participating in Agile and Scrum development workflows.
- Collaborated with team of 6 members using version control, code reviews, and collaborative development practices.
- Tech Stack: Android Studio, Java, GitHub, Discord, WebEx, Trello

3D Escape Room Simulator

Jan 2019 – May 2019

Pleasant Valley High School

Bettendorf, IA

- Created 3D game assets, terrain, and level designs for escape room simulator game.
- Implemented randomized gameplay logic for procedural level generation and puzzle variation.
- Tech Stack: C#, Unity Game Engine, Blender 3D

Education

Bachelor of Science — BS, Software Engineering

2020 – 2025

Iowa State University

High School Diploma

2016 – 2020

Pleasant Valley High School

Certifications & Achievements

- Apple Certified Repair Technician
- Dell Certified Repair Technician
- C++ Class Development
- C++ Programming for Unreal Game Development
- Intermediate Object-Oriented Programming for Unreal Games
- Introduction to C++ Programming and Unreal
- More C++ Programming and Unreal
- Unreal Engine 5 C++ — The Ultimate Game Developer
- Microsoft Technology Associate – Introduction to Programming with Java
- Rank #1 for C++ Programming on HackerRank